NATIONAL UNIVERSITY OF MODERN LANGUAGES

Lab no 4

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PROGRAM: BSSE (6th Semester)

COURSE: Software Quality Engineering Lab

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Task #1

Mapping of Project Quality Requirements to Two Product Quality Models

Project Title: **KAAMKER – AI-Enabled Local Services Marketplace**

Selected Quality Models:

1. **FURPS Model**
2. **ISO/IEC 25010:2011 Model**

The quality requirements of the KaamKer system can be mapped to the following two software quality models.

1. FURPS Quality Model

The **FURPS model** is a software quality framework that categorizes system requirements into five major areas:

- Functionality
- Usability
- Reliability
- Performance
- Supportability

These categories help evaluate both functional and non-functional aspects of a software system.

Functionality

Functionality refers to the core capabilities of the system and whether the software performs the tasks required by users.

In the **KaamKer platform**, functionality includes:

- User registration and login
- Worker profile creation and management
- Service listing by workers
- Job posting and booking requests
- Messaging between customers and workers
- Rating and review system
- AI-based worker ranking
- Fake service detection mechanism

These features ensure that the platform successfully connects customers with skilled workers.

Usability

Usability measures how easy and convenient the system is for users.

For the **KaamKer platform**, usability will be ensured through:

- A simple and intuitive mobile interface
- Easy navigation between different services
- Clear service categories
- Simple booking workflow

The goal is to make the application easy to use for both customers and workers.

Reliability

Reliability refers to the system's ability to perform consistently without failures.

For KaamKer, reliability includes:

- Accurate processing of booking requests

- Reliable messaging between users
- Secure storage of ratings and reviews
- Stable system performance without frequent crashes

A reliable system builds trust among users.

Performance

Performance refers to the system's speed, responsiveness, and ability to handle multiple users.
For KaamKer, performance requirements include:

- Fast search results for services
- Efficient processing of booking requests
- Quick response from the AI ranking algorithm
- Ability to support many users simultaneously

Good performance ensures a smooth user experience.

Supportability

Supportability refers to how easily the system can be maintained, updated, and improved in the future.
KaamKer supports maintainability through:

- Modular system architecture
- API-based backend services
- Cloud integration for scalability
- Easy integration of new features in future versions

This ensures long-term sustainability of the system.

2. ISO/IEC 25010:2011 Quality Model

The **ISO/IEC 25010:2011** model is an international standard for evaluating software product quality. It defines several quality characteristics that help measure software effectiveness.
Key characteristics relevant to KaamKer include:

- Functional Suitability
- Performance Efficiency
- Usability
- Reliability
- Security
- Maintainability
- Portability

Functional Suitability

Functional suitability evaluates whether the system provides functions that meet user needs.
KaamKer supports functional suitability through:

- Worker discovery and search
- Service booking system
- Rating and review mechanism
- AI-based worker recommendation system

These features ensure the system fulfills its intended purpose.

Performance Efficiency

Performance efficiency measures how effectively the system uses resources and delivers fast results.

In KaamKer:

- Search results should load quickly
- Booking processes should execute efficiently
- AI ranking should generate results with minimal delay

Efficient performance improves user satisfaction.

Usability

Usability ensures the system is easy to learn and use.

KaamKer will focus on:

- Simple user interface design
- Clear instructions and navigation
- Easy service search and booking process

This makes the platform accessible to both customers and workers.

Reliability

Reliability ensures the system operates consistently over time.

For KaamKer:

- Booking data must remain accurate
- Messaging system must function properly
- AI ranking results must remain stable

Reliable systems increase user confidence in the platform.

Security

Security protects user data and prevents unauthorized access.

Security features for KaamKer include:

- Secure authentication for users
- Protection of personal data
- Secure database management
- Monitoring for fake or suspicious services

Security is essential for maintaining user trust.

Maintainability

Maintainability refers to how easily the system can be modified or improved.

KaamKer will support maintainability by:

- Using modular software design
- Implementing well-structured APIs
- Maintaining clear documentation for future updates

This allows developers to update the system easily.

Portability

Portability refers to the ability of the software to run on different platforms.

KaamKer will be developed using **Flutter**, which allows the application to run on multiple platforms such as:

- Android
- iOS

This increases the accessibility and reach of the platform.

Task #2

Justification for Selecting the Quality Models

Justification for Selecting the FURPS Model

The **FURPS model** was selected because it provides a clear and simple structure for evaluating software quality.

Reasons for selecting FURPS:

- It focuses on both functional and non-functional requirements.
- It provides a simple classification of system quality attributes.
- It is widely used in software engineering projects.
- It helps evaluate usability, performance, and reliability effectively.

Since KaamKer is a marketplace platform that requires strong functionality and performance, the FURPS model is suitable for evaluating its quality.

Justification for Selecting the ISO/IEC 25010 Model

The **ISO/IEC 25010:2011 model** was selected because it is an internationally recognized standard for software quality evaluation.

Reasons for selecting ISO 25010:

- It provides a comprehensive software quality framework.
- It includes important characteristics such as security, maintainability, and portability.
- It is suitable for modern systems that include mobile applications and AI components.
- It is widely used in industry and academic research.

Since KaamKer integrates **mobile technology, cloud services, and AI-based features**, ISO 25010 provides a detailed framework to evaluate the overall quality of the system.